

WEEK – 5 DBMS

Payment Transaction Management System

Description

1. Create Payment table.
2. Store payment details such as mode, date, and status.
3. Manage successful and failed transactions.
4. Analyze payment methods used by customers.
5. Generate payment transaction reports.

1. Create Payment Table

```
CREATE TABLE Payment (  
    Payment_ID INT PRIMARY KEY,  
    Customer_ID INT,  
    Payment_Mode VARCHAR(20),  
    Payment_Date DATE,  
    Amount DECIMAL(10,2),  
    Payment_Status VARCHAR(20)  
);
```

2. Insert Payment Details

```
INSERT INTO Payment  
VALUES  
(101, 1, 'UPI', '2026-09-01', 1500.00, 'Successful'),  
(102, 2, 'Card', '2026-09-02', 2500.00, 'Successful'),  
(103, 3, 'Cash', '2026-09-02', 1000.00, 'Successful'),  
(104, 4, 'UPI', '2026-09-03', 1800.00, 'Failed'),  
(105, 5, 'Card', '2026-09-04', 3200.00, 'Successful');
```

3. Display Successful Transactions

```
SELECT * FROM Payment  
WHERE Payment_Status = 'Successful';
```

4. Display Failed Transactions

```
SELECT * FROM Payment  
WHERE Payment_Status = 'Failed';
```

5. Analyze Payment Methods

```
SELECT Payment_Mode, COUNT(*) AS Total_Transactions  
FROM Payment  
GROUP BY Payment_Mode;
```

6. Calculate Total Successful Payment Amount

```
SELECT SUM(Amount) AS Total_Successful_Amount  
FROM Payment  
WHERE Payment_Status = 'Successful';
```

7. Generate Payment Transaction Report

```
SELECT Payment_Mode,  
    Payment_Status,  
    COUNT(*) AS Total_Transactions,
```

```
        SUM(Amount) AS Total_Amount  
FROM Payment  
GROUP BY Payment_Mode, Payment_Status;
```

Conclusion

The Payment Transaction Management System stores payment information, tracks successful and failed transactions, analyzes payment methods, and generates transaction reports using SQL.